

Biodiversity risk and agricultural futures markets

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ABSTRACT

This study constructs a novel Biodiversity Risk (BR) index based on textual analysis and evaluates its performance in forecasting volatility in China's major agricultural futures markets. Using an autoregressive (AR) model framework, we conduct comparative analyses of the BR index's predictive ability through univariate models, various combination forecasting methods, and dimensionality reduction techniques (PCA, sPCA, and PLS). Empirical results indicate that the BR index possesses a certain degree of forecasting power for volatility in agricultural futures markets, with PLS-based models demonstrating relatively superior predictive performance across multiple markets. This conclusion is supported by multiple robustness tests, high-volatility periods, and medium- to long-term forecasts. The study emphasizes that incorporating biodiversity risk indicators can help improve risk management in agricultural futures and promote sustainable development.

1. Introduction

The agricultural futures market in China is regarded as playing an essential role in ensuring national food security and maintain economic stability, as fluctuations in agricultural commodity prices have a profound impact on the macroeconomy (Li, Li, Liu, Zhu, & Wei, 2022). This market plays a crucial role in price discovery and risk management, allowing stakeholders to hedge against price volatility and manage uncertainties in supply and demand. In recent years, volatility in agricultural futures markets has increased significantly due to global health crises and escalating geopolitical tensions, which have heightened uncertainties across global food supply chains (Lu et al., 2025). These external factors have exacerbated market instability, highlighting the necessity for forecasting methods and risk control strategies (Luo, Klein, Ji, & Hou, 2022).

Volatility, a principal indicator of market uncertainty, measures price variation and market risk (Gong & Lin, 2021). Increased volatility in agricultural commodity prices can undermine food security, elevate production risks, and disrupt supply chain planning (Li et al., 2022; Ma, Lu, Wang, & Chevallier, 2021). In recent years, researchers have increasingly focused on the volatility of agricultural futures markets, focusing not only on internal market mechanisms but also on external influences such as macroeconomic shocks, institutional interventions, and climatic disturbances (He & Serra, 2022; Lu et al., 2025). A substantial body of literature has focused on estimating the overall volatility

dynamics of futures markets, particularly for crude oil (Salisu, Gupta, & Demir, 2022; Zhang, Wahab, & Wang, 2023) and energy markets (Chen, Bao, & Jin, 2021; Wang, Zhao, Liang, & Jiu, 2022; Zhang, Xiang, Zou, & Guo, 2024). Research in agricultural futures markets has predominantly focused on the interconnections between agricultural and other futures markets, particularly with price linkages and spillover effects (Lucotte, 2016; Sun, Mirza, Qadeer, & Hsueh, 2021; Tiwari, Abakah, Adewuyi, & Lee, 2022). These findings underscore the necessity for more robust and responsive tools to assess and manage volatility within commodity-based risk management systems (Luo et al., 2022). He and Serra (2022) examined the effects of institutional actions, such as price controls, whereas Li et al. (2022) assessed the impact of policy and weather-related news through text mining and sentiment analysis. Khalfaoui, Goodell, Mefteh-Wali, Chishti, and Gozgor (2024) have investigated the dynamic effects of climate change on global agriculture and food markets utilizing time-frequency approaches.

Although existing studies have widely examined agricultural futures volatility in relation to macroeconomic shocks, policy uncertainty, and climate risks, ecological factors such as biodiversity loss have received relatively little attention as potential sources of market volatility.

Biodiversity is a vital component of the trade-off between environment and agriculture system. It supports essential ecosystem services such as pollination, water regulation, and pest control, all of which are indispensable for stable food production. The deterioration or elimination of these services could significantly jeopardize food production

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1057-5219/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(Ortiz, Outhwaite, Dalin, & Newbold, 2021). Human activities and extreme weather events have profoundly affected biodiversity (Butt et al., 2013; Legge, Rumpff, Garnett, & Woinarski, 2023; Pörtner et al., 2023). Biodiversity risk refers to the uncertainty introduced into agricultural systems due to ecological degradation, species extinction, and imbalances within ecosystems. This form of risk has emerged as a significant external shock to agricultural commodity markets. As awareness of the economic consequences of biodiversity loss continues to grow, financial markets have begun to incorporate biodiversity-related factors into broader environmental risk assessments (Carvalho, Cojoiianu, & Ascui, 2023). Recognizing biodiversity as a material issue for both environmental and financial systems has led to the development of a new research agenda within biodiversity-related finance. This emerging literature seeks to evaluate the impact of biodiversity risk on asset valuation, firm performance, and overall market stability (Adamolekun, 2024; Giglio, Kuchler, Stroebel, & Zeng, 2023).

In recent years, scholars have increasingly examined the financial dimensions of biodiversity loss, especially through the role of private capital in addressing ecological degradation and promoting sustainable investment (Karolyi & Tobin-de la Puente, 2023; Flammer, Giroux, & Heal, 2025; Panwar, Ober, & Pinkse, 2023; Carvalho et al., 2023; Garel, Romec, Sautner, & Wagner, 2024; Coqueret, Giroux, & Zerbib, 2025). However, many existing studies embed biodiversity risk within broader ESG or environmental frameworks without isolating its specific impact on price volatility and trading behavior (Kopnina, Zhang, Anthony, Hassan, & Maroun, 2024; Zhu & Carrasco, 2025). Empirical models tend to rely on firm-level or macro-level indicators, with limited integration of ecological or biodiversity-specific data (Bach, Hoang, & Le, 2025; Bassen, Buchholz, Lopatta, & Rudolf, 2024), thereby constraining our understanding of market responses to biodiversity-related shocks. For example, biodiversity risk has been shown to adversely affect firm performance, particularly in biodiversity-sensitive industries, by reducing profitability and increasing production costs (Bach et al., 2025). Moreover, investors are responsive to biodiversity-related policy signals, with firms that proactively manage such risks experiencing positive abnormal stock returns following key biodiversity events (Kalhor & Kyaw, 2024), suggesting that markets perceive biodiversity management as economically valuable. At the corporate finance level, biodiversity risk prompts firms to increase cash holdings as a precautionary measure, especially under high competition and financial constraints (Ahmad & Karpuz, 2024). Such behavior reflects a broader market perception of biodiversity risk as a source of uncertainty. Taken together, these findings confirm that although biodiversity risk originates from ecological factors, it has concrete financial consequences, affecting asset prices, firm strategies, and investor behavior.

Despite the growing recognition of biodiversity's financial implications (Carvalho et al., 2023), empirical research directly linking biodiversity-related risks to commodity price dynamics, volatility, and investor behavior in agricultural futures markets remains limited. This research gap is particularly significant given the strong dependence of agricultural production on ecological stability. While existing studies often focus on external shocks such as climate events, macroeconomic fluctuations, and policy interventions (Pettorelli et al., 2021; Smith et al., 2022; Trew & Maclean, 2021), biodiversity loss is frequently overlooked as a persistent and systemic source of market uncertainty.

While biodiversity risk has gained increasing attention in financial research, its direct role in shaping commodity market price dynamics, especially in agricultural futures markets, remains underexplored. To address this gap, the present study focuses on the intersection between biodiversity risk and volatility in commodity markets.

The empirical approach consists of the following steps. Step one involves constructing the Biodiversity Risk (BR) index using text analysis methods based on news content from four major media outlets: Guangming Daily, Economic Daily, People's Daily, and CCTV. On this basis, we establish a benchmark forecasting framework by employing an autoregressive (AR) model to predict volatility in agricultural futures

markets. To enhance model performance, we further incorporate five combination forecasting methods (f_{MEAN} , f_{MEDIAN} , f_{TMC} , f_{DMSPE100} , and f_{DMSPE90}) and three dimensionality reduction techniques (PCA, sPCA, and PLS). The results show that the BR index possesses significant predictive power for volatility in these markets. The second step compares the BR-based models with conventional uncertainty indicators (TPU, EPU, CNEPU, GPR, CCPU). The findings indicate that uncertainty and geopolitical risk indicators are typically less efficacious than the AR-BRPLS model in forecasting volatility in principal agricultural futures markets. Step three performs multiple robustness tests to verify the results, including rolling window estimation, alternative out-of-sample forecast horizons, and reconstructing the BR index to assess the stability of its construction process. Step four examines supplementary attributes of the forecasting framework via comprehensive evaluations, encompassing out-of-sample forecasting efficiency in high-volatility environments and across various forecast timeframes.

This research provides three significant contributions to the existing literature: This study presents an innovative biodiversity risk indicator. Research on biodiversity risk and the formulation of associated indicators has been progressively intensifying. An increasing number of researchers have started to develop biodiversity risk indices to measure market focus and evaluate financial consequences. Giglio et al. (2023) initiated the news-based macro-level biodiversity risk index and subsequently formulated firm-level exposure metrics utilizing 10-K reports, survey data, and biodiversity fund holdings. Ma, Wu, and Zeng (2024) introduced a comprehensive Biodiversity Risk (BR) index that assesses both physical and transition hazards, illustrating its predictive capability for market risk premia and its utility in asset allocation. Our work develops a novel BR index based on biodiversity-related news from four prominent media outlets. In accordance with Ardia et al. (2023), Facchini, Matin, and Skiadopoulos (2023), and Garel et al. (2024), we augment a collection of biodiversity-related seed terms utilizing the Word2Vec algorithm and produce the index via text analysis. This method allows us to quantify the intensity of biodiversity risk awareness throughout time.

This study emphasizes the predictive capability of the Biodiversity Risk (BR) index by comparing it to the Geopolitical Risk (GPR) index and several uncertainty indices, while also affirming its forecasting accuracy under high-volatility conditions and across different time horizons. Furthermore, of the forecasting models assessed, the Partial Least Squares (PLS) model demonstrates the greatest overall accuracy when consolidated across various agricultural futures markets.

The organization of the subsequent sections of this paper is as follows: Section 2 presents the data utilized for the empirical investigation. Section 3 delineates the forecasting models, and the evaluation methodologies utilized. Section 4 presents the out-of-sample forecasting results, encompassing performance comparisons among geopolitical risk and uncertainty indices, along with forecast encompassing tests. Section 5 performs multiple robustness checks to corroborate the findings. Section 6 provides further analysis, highlighting out-of-sample forecasting precision during periods of high volatility and across multiple forecasting horizons. Section 7 presents the major findings and conclusions of the paper.

2. Data description

2.1. Biodiversity risk index

This study examines the predictability of the Biodiversity Risk index (BR) of volatility in agricultural futures markets. The dataset includes the BR index and five uncertainty indices. Due to the intricate and subtle characteristics of biodiversity-related risks, conventional econometric techniques are inadequate for precise measurement. To tackle this issue, we employ a text-based methodology to develop the BR index, utilizing four primary news outlets: (1) China Central Television News, (2)

People's Daily, (3) Guangming Daily, and (4) Economic Daily. The rationale resides in their unmatched influence and policy-making roles in China. China Central Television News is a national television news channel that offers official policy interpretations and comprehensive coverage, possessing the utmost authority in public discourse. In comparison to other news outlets, People's Daily, Guangming Daily, and Economic Daily are closely associated with the central government, therefore wielding far higher influence. Since its establishment, the People's Daily has functioned as the official publication of the Central Committee of the Chinese Communist Party. Guangming Daily exerts influence in the domains of research, education, and culture. Economic Daily focuses on macroeconomic analysis and discussion on industrial policy. The initial stage in developing the biodiversity risk index is to identify material pertinent to biodiversity. We commence with seven seed keywords: Biodiversity, Genetic Diversity, Species Diversity, Ecosystem Diversity, Ecological Protection, Biosphere, and Biodiversity Loss. Employing the Word2Vec methodology, we augment these seed keywords to encompass semantically associated words, thereafter doing manual verification to guarantee relevance and accuracy. We develop a biodiversity dictionary through this technique. Table 1 displays a total of 35 keywords, categorized into seven subject groups.

Utilizing the dictionary, we calculate the frequency of relevant terms inside each article. Articles in which biodiversity-related terms appear more than once are classified as biodiversity-related. Biodiversity risk is defined as the degree to which materials convey unfavorable perceptions or apprehensions regarding biodiversity and its accompanying threats. We analyze the sentiment and risk orientation present in the identified articles. We utilize the sentiment dictionary created by Jiang, Lee, Martin, and Zhou (2019) to differentiate between positive and negative terms and apply the risk lexicon derived from Linguistic Inquiry and Word Count (LIWC) as described by Kim (2013) to detect risk-related language. This establishes the foundation for the Biodiversity Risk Index (BR), which is calculated as follows:

Table 1
Construction of the biodiversity dictionary.

Variables	Category	Keywords
Biodiversity	Biodiversity	Ecological Diversity Biodiversity Conservation Ecosystem Functioning Ecological Environment
	Genetic Diversity	Gene Diversity Genetic Structure Population Genome
	Species Diversity	Plant Diversity Community Structure Ecosystem Services
Ecosystem Diversity	Ecosystem Diversity	Biological Species Species Richness Biological Habitat
		Ecological Environment Protection Ecological Civilization Development Wetland Conservation Ecological Governance Resource Conservation Water Resource Conservation Ecological Restoration
		Biosphere
Ecological Protection	Ecological Protection	Ecosystem Biological Chain Material Cycle Biosphere/Biotic World Food Chain
		Biodiversity Loss
		Forest Degradation Deforestation Land Degradation Ocean Acidification Ecosystem Collapse

Notes: This table presents the biodiversity lexicon, which includes seven categories of seed words along with their corresponding 35 keywords.

$$BR_{i,t} = 100 \times \left(\frac{\text{Negative}_{i,t} - \text{Positive}_{i,t}}{\text{Negative}_{i,t} + \text{Positive}_{i,t}} + 1 \right) * \frac{1}{2} * \frac{\text{Risk}_{i,t}}{\text{Length}_{i,t}}, \quad (1)$$

where $BR_{i,t}$ represents the biodiversity risk of the i -th article at time t , $\text{Positive}_{i,t}$ and $\text{Negative}_{i,t}$ indicate the counts of positive and negative sentiment words, respectively, $\text{Risk}_{i,t}$ is the number of risk-related terms; and $\text{Length}_{i,t}$ represents the total number of words in the i -th article. $\frac{\text{Risk}_{i,t}}{\text{Length}_{i,t}}$ measures the percentage of risk-related words in the text, accounting for differences in article length by calculating this percentage.

$\left(\frac{\text{Negative}_{i,t} - \text{Positive}_{i,t}}{\text{Negative}_{i,t} + \text{Positive}_{i,t}} + 1 \right) * \frac{1}{2}$ measures the sentiment of the document, with values ranging from 0 to 1, where a higher value indicates a greater degree of negativity in the article.

To obtain the monthly biodiversity risk index, we average the daily BR values within each month:

$$BR_{i,m} = \frac{\sum_{n=1}^{N_m} BR_{i,t}}{N_m}, \quad (2)$$

where N_m is the number of days in month m .

2.2. Other variables

Other variables comprise the U.S. Economic Policy Uncertainty (EPU_US), the Geopolitical Risk Index (GPR),¹ and indices such as the China Trade Policy Uncertainty Indices Based on Mainland Papers (TPU_M), the China Economic Policy Uncertainty Index (CNEPU), and the Chinese climate policy uncertainty (CCPU). The CCPU data is sourced from the research conducted by Ma et al. (2023).²

3. Methodology

3.1. Realized variance

In this study, we utilize intraday returns of agricultural futures to construct monthly realized variance (RV). Methodological frameworks that estimate regression analysis for conditional volatility estimation invariably utilize ex-post variance measures constructed from historical data. Following Schwert (1989) and Paye (2012), the realized volatility for month t is calculated as the sum of the squared daily returns within that month:

$$RV_t = \sum_{j=1}^M r_{t,j}^2, \quad (3)$$

where $r_{t,j}$ represents the j -th daily return of month t and M denotes the number of observations of month t .

Although the realized variance (RV) computed using eq. (1) is a standard measure, it generally exhibits non-normality. In our volatility forecasting regressions, we estimate coefficients employing the ordinary least squares (OLS) method. Nevertheless, inference relies on OLS estimations. To mitigate this issue, following Paye (2012), we apply a natural logarithmic transformation to the RV ($RV_t = \log(RV_t)$), which yields a distribution that more closely approximates normality.

3.2. Predictive model

This section provides a comprehensive overview of the models and methodologies employed in this study, which are categorized into three

¹ EPU_US, GPR, TPU_M, CNEPU data can be downloaded from: https://www.policyuncertainty.com/global_monthly.html

² CCPU data can be downloaded from: <https://figshare.com/>

main types: (1) the autoregressive (AR) model and its univariate extensions, (2) AR-based models integrated with dimensionality reduction techniques. (3) a combination forecasting framework.

We integrate dimensionality reduction techniques (PCA, sPCA, and PLS) into the AR framework to address potential multicollinearity, reduce overfitting risks, and extract the most informative components from multiple biodiversity-related indices. To further enhance forecast robustness and mitigate model uncertainty, we adopt five ensemble approaches: f_MEAN, f_MEDIAN, f_TMC, f_DMSPE100, and f_DMSPE90.

3.2.1. Autoregressive (AR) framework

The autoregressive (AR) model is a fundamental instrument in time series analysis and often acts as a benchmark for forecasting applications. Due to its simplicity, explanatory power, and efficiency in capturing temporal dependencies, the AR model has been extensively utilized to forecast financial and economic variables, including asset returns, exchange rates, and interest rates (Christiansen, Schmeling, & Schrimpf, 2012; Ma, Liu, Wahab, & Zhang, 2018; Tang & Zhong, 2023; Wang, Wei, Wu, & Yin, 2018). This study used the AR model as the benchmark forecasting model, aligning with prior empirical research. The formal specification of the AR model is presented as follows, in accordance with the methodologies of Ma et al. (2018), Guo, He, Liang, and Ma (2022), and Tang and Zhong (2023).

$$RV_{t+1} = \alpha_0 + \rho_1 RV_t + \varepsilon_{t+1}, \quad (4)$$

where α_0 is a constant term, and ε_{t+1} denotes the error term.

To assess the predictive power of the BR index for the volatility of agricultural commodity futures markets, we extend the baseline AR(1) model by incorporating each predictor individually. These include the BR^{CCTV} , BR^{PEO} , BR^{GM} , and BR^{ECO} indices. The extended model, referred to as the AR-BR model, is specified as:

$$RV_{t+1} = \alpha_0 + \rho_1 RV_t + \beta_i X_{i,t}^m + \varepsilon_{t+1}, \quad (5)$$

where m denotes the individual indicators, which include four biodiversity indices constructed from four news media sources (BR^{CCTV} , BR^{PEO} , BR^{GM} , and BR^{ECO}).

3.2.2. PCA factor in the AR model

Principal Component Analysis (PCA) is a classic linear dimensionality reduction technique that transforms correlated variables into uncorrelated principal components, ordered by the amount of variance they explain (Chen & Han, 2021; Jiang, Tang, & Zhou, 2018; Pearson, 1901). To capture the common predictive information embedded in multiple BR index, we integrate PCA into the AR(1) framework, resulting in the AR-PCA model for volatility forecasting. The procedure is defined as follows:

$$x_{i,t} = \lambda_i' F_{PCA,t} + e_{i,t}, \quad (6)$$

where $x_{i,t}$ denotes the standardized variables of the i -th indicator in month t , and $F_{PCA,t}$ represents the principal component extracted by PCA. This process serves as the foundation for constructing an AR model that leverages shared information across indicators.

We apply PCA to the BR^{CCTV} , BR^{PEO} , BR^{GM} , and BR^{ECO} indicators to incorporate the predictive information. These components are incorporated into an AR(1), the general form of the AR-PCA model is as follows:

$$RV_{t+1} = \alpha_0 + \rho_1 RV_t + \beta_1 F_{PCA,t} + \varepsilon_{t+1}, \quad (7)$$

where F_{PCA} represents the first principal component extracted from the four biodiversity risk indices. Following Stock and Watson (2004), we use the first principal component, which explains the majority of the variance in the data, to effectively reduce dimensionality while preserving key information.

3.3. sPCA factor in the AR model

scaled Principal Component Analysis (sPCA) is a dimensionality reduction technique. Unlike traditional principal component analysis (PCA), which identifies components by maximizing the shared variance among predictors, sPCA enhances forecasting accuracy by scaling each predictor according to its estimated predictive slope with respect to the target variable. (Huang, Jiang, Li, Tong, & Zhou, 2022). The general procedure is as follows:

$$RV_{t+1} = \alpha_0 + \beta_i X_{i,t} + \varepsilon_{t+1}, \quad (8)$$

where β_i reflects the normalized weight of each indicator, we proceed to compute their principal components ($\beta_1 X_{1,t}, \beta_2 X_{2,t}, \dots, \beta_i X_{i,t}$):

$$\beta_i X_{i,t} = \lambda_i' F_{sPCA,t} + e_{i,t}, \quad (9)$$

where $F_{sPCA,t}$ denotes the comprehensive index constructed from all biodiversity risk measures using the sPCA approach. Principal components are extracted from the BR^{CCTV} , BR^{PEO} , BR^{GM} , and BR^{ECO} indices using this approach. The general AR-sPCA model is presented as follows:

$$RV_{t+1} = \alpha_0 + \rho_1 RV_t + \beta_1 F_{sPCA,t} + \varepsilon_{t+1}, \quad (10)$$

3.3.1. PLS factor in the AR model

Proposed by Wold (1966), partial least squares (PLS) is a supervised approach that integrates regression analysis with factor extraction for dimensionality reduction. It enables the identification of latent components from multiple predictors while filtering out irrelevant variations. Unlike unsupervised methods, PLS explicitly accounts for the relationship between predictors and the target variable, making it particularly effective for forecasting. The procedure consists of two steps. First, underlying factors are extracted based on the ability of the BR index to explain future realized volatility:

$$X_{i,t-1} = \alpha_{i,0} + \beta_i RV_t + e_{i,t-1}. \quad (11)$$

Second, the PLS common index is derived by regressing each variable $X_{i,t}$, on its estimated coefficient $\hat{\beta}_i$:

$$X_{i,t} = \varphi_{i,0} + \hat{\beta}_i F_{PLS,t} + \mu_{i,t-1}, \quad (12)$$

where $F_{PLS,t}$ denotes the common index extracted by the PLS, the PLS model can be defined as:

$$RV_{t+1} = \alpha_0 + \rho_1 RV_t + \beta_1 F_{PLS,t} + \varepsilon_{t+1}, \quad (13)$$

The construction of all predictive models employed in this study is comprehensively described in this section. For further details, please refer to Table 3.

3.4. Evaluation method

To evaluate the predictive performance of the forecasting models, we adopt the out-of-sample R^2 as the primary evaluation criterion, following Campbell and Thompson (2008) and Rapach, Strauss, and Zhou (2010). This metric measures the percentage reduction in mean squared prediction error (MSPE) of a forecast model relative to a benchmark model using data not employed during model estimation. The R_{oos}^2 statistic can be expressed as:

$$R_{oos}^2 = 1 - \frac{\sum_{t=M+1}^T (RV_t - \widehat{RV}_{t,F})^2}{\sum_{t=M+1}^T (RV_t - \widehat{RV}_{t,B})^2}, \quad (14)$$

where T represents the length of the out-of-sample period, and M is the number of in-sample observations. RV_t represents the realized volatility at time t , $\widehat{RV}_{t,B}$ is the forecast from the benchmark AR(1) model, and $\widehat{RV}_{t,F}$ denotes the forecast from the alternative predictive models. A

positive R^2_{os} indicates that the forecasting model outperforms the benchmark by achieving a lower MSPE, demonstrating superior predictive capability.

Furthermore, to assess whether the observed improvement in predictive accuracy is statistically significant, we employ the MSPE-adjusted test proposed by Clark and West (2007). This test evaluates whether the reduction in MSPE achieved by the forecasting model relative to the benchmark is statistically meaningful, and is formally defined as follows:

$$f_t = (RV_t - \widehat{RV}_{t,B})^2 - (RV_t - \widehat{RV}_{t,F})^2 + (\widehat{RV}_{t,B} - \widehat{RV}_{t,F})^2. \quad (15)$$

4. Empirical results

4.1. Descriptive statistics

Fig. 1 illustrates the RV of the agricultural futures market from January 2010 to December 2023. Pronounced volatility peaks are observed around 2016 and 2020 across multiple commodities, corresponding to elevated global market uncertainty (Brogaard, Dai, Ngo, & Zhang, 2020; Uddin, Chowdhury, Anderson, & Chaudhuri, 2021). Specifically, corn and common wheat experienced sharp increases in 2016 (Zhao & Ju, 2024). In contrast, palm oil experienced a sustained increase in volatility beginning in 2020, likely attributable to the COVID-19 pandemic and ensuing supply chain disruptions (Van Hoek, 2020).

Fig. 2 illustrates the time series of biodiversity risk indices (BR) constructed from four major Chinese media sources: Guangming Daily, Economic Daily, People's Daily, and CCTV, covering the period from January 2010 to December 2023. Across all sources, the BR indices exhibit noticeable temporal variation, with several periods marked by significant surges.

Table 2 presents the descriptive statistics for biodiversity risk from four news sources, RV from six agricultural commodity futures markets, and the other variables. First, among the four biodiversity risk indices constructed from different news sources, BR^{CCTV} exhibits the lowest mean and standard deviation, indicating relatively high stability. In contrast, BR^{PEO} shows the highest mean. Second, all variables have positive skewness, suggesting that the distributions are right-skewed. Third, most variables display kurtosis greater than zero, indicating a leptokurtic nature, especially the RVs across various markets, which exhibit pronounced peaks. However, the kurtosis values for BR^{ECO} and

CCPU are slightly below those of a normal distribution. Finally, the augmented Dickey-Fuller (ADF) test rejects the null hypothesis of a unit root at the 1 % significance level, confirming that the data series are stationary and thus suitable for direct modeling.

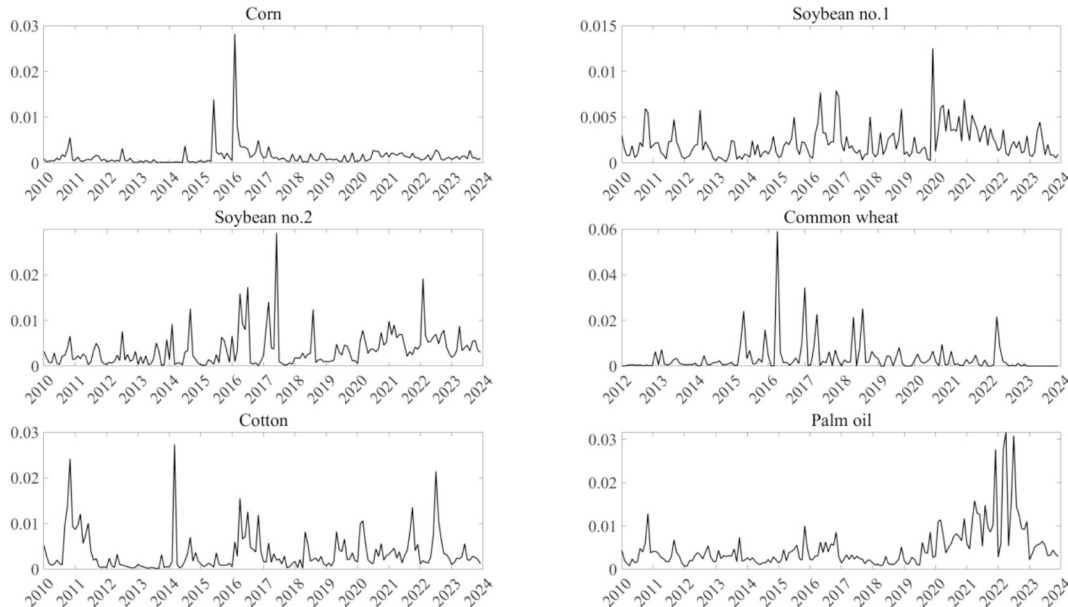
4.2. Out-of-sample forecasting results

Out-of-sample forecasting performance is a critical measure of a model's predictive ability. Here, we examine the out-of-sample predictive power of the Biodiversity Risk Index in forecasting volatility in agricultural commodity futures markets. The full sample covers 168 monthly observations from January 2010 to December 2023. We set the in-sample period as January 2010 through December 2019 (120 months) and the out-of-sample period as January 2020 through December 2023 (48 months).

Table 4 reports the autoregressive (AR) model regression results for the risk indices constructed from four news sources. The predictive power for soybean futures is relatively weaker, likely due to China's heavy reliance on soybean imports. Its price and volatility are mainly influenced by global factors such as trade tensions, exchange rates, and U.S. policies rather than domestic ecological or sentiment indicators (Lu et al., 2025). Consequently, domestic news-based biodiversity risk signals are less effective in capturing soybean price dynamics. In contrast, commodities like wheat and corn, which are more domestically produced, respond more directly to national biodiversity concerns. Additionally, soybean prices show strong seasonal patterns (Gouel, 2020) and are shaped by long-term global contracts, further limiting the forecasting ability of short-term biodiversity sentiment indicators.

Table 5 supplements a comparison of forecasting performance for indicators such as geopolitical risk, thereby supporting this notion. We first focus on the predictive performance of each Biodiversity Risk (BR) series across the remaining four markets.

As shown in Table 4, among the individual-variable models based on different BR sources, BR^{GM} exhibited superior performance for multiple contracts. Specifically, its out-of-sample R^2 for corn reached 12.058 % and was statistically significant at the 5 % level, the highest among all single-predictor models. The R^2_{os} for Soybean No. 2 was 5.620 %, significant at the 1 % level. BR^{GM} also delivered strong predictive results for cotton and palm oil. In contrast, BR^{ECO} and BR^{PEO} generally yielded weaker performance, with negative out-of-sample R^2 for most commodities observed for BR^{PEO} . Combined forecasting methods produced



Figs. 1. Agricultural futures realized volatility during the whole sample period.

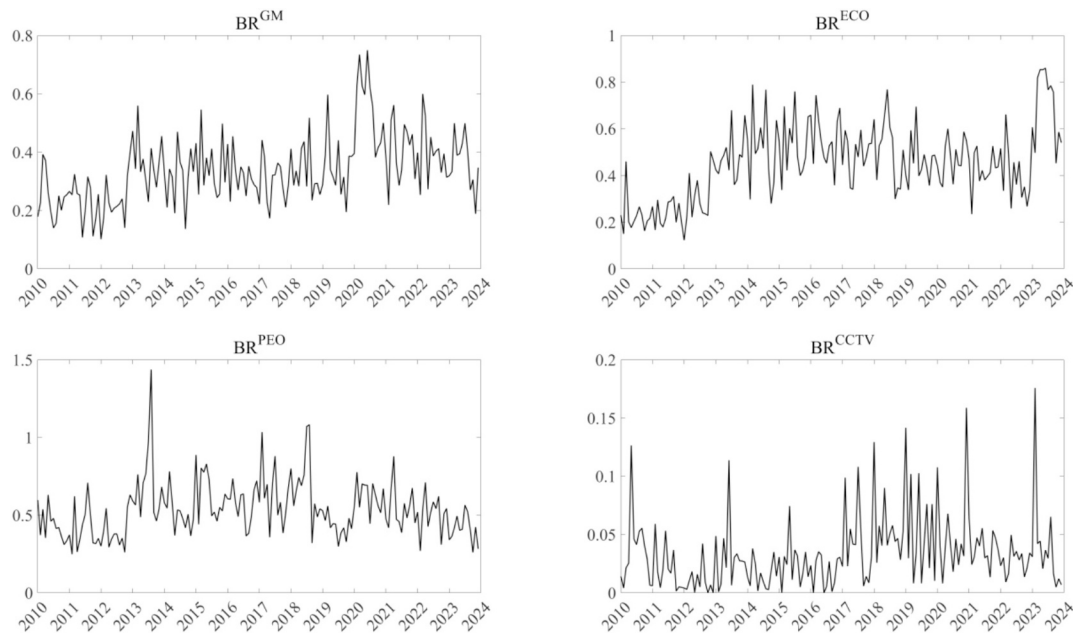


Fig. 2. Biodiversity Risk index(BR) during the whole sample period.

Table 2
Summary statistics of all data.

	mean	SD	Skewness	Kurtosis	ADF	Period
RV_corn	0.001	0.003	7.903	76.849	−10.238***	January 2010–December 2023
RV_so1	0.002	0.002	2.067	6.525	−9.329***	January 2010–December 2023
RV_so2	0.004	0.004	2.958	13.409	−10.024***	January 2010–December 2023
RV_cw	0.003	0.007	4.764	29.199	−11.013***	January 2012–December 2023
RV_cotton	0.004	0.004	2.820	10.338	−8.194***	January 2010–December 2023
RV_palm	0.005	0.005	3.104	11.974	−6.921***	January 2010–December 2023
BR ^{GM}	0.340	0.120	0.727	0.790	−7.791***	January 2010–December 2023
BR ^{ECO}	0.452	0.164	0.286	−0.288	−6.745***	January 2010–December 2023
BR ^{PEO}	0.537	0.178	1.310	3.678	−8.325***	January 2010–December 2023
BR ^{CCTV}	0.034	0.031	1.981	5.029	−11.674***	January 2010–December 2023
TPU_M	233.907	243.194	2.063	5.432	−6.088***	January 2010–December 2023
EPU_US	162.105	67.653	2.038	6.142	−5.393***	January 2010–December 2023
CNEPU	145.331	28.218	1.000	1.111	−7.006***	January 2010–December 2023
GPR	98.490	30.112	3.238	18.087	−5.682***	January 2010–December 2023
CCPU	2.348	0.538	0.554	−0.172	−6.861***	January 2010–December 2023

Notes: This table presents descriptive statistics for realized volatilities of six agricultural commodity stock returns, four-source Biodiversity Risk indices, four sub-indices of Uncertainty indices and Geopolitical Risk (GPR) Index. For each variable, the mean, standard deviation (SD), skewness, kurtosis, and the Augmented Dickey–Fuller (ADF) test statistic are reported. The ADF test evaluates the null hypothesis of a unit root (i.e., non-stationarity). The asterisk *** indicates rejection of the null hypothesis at the 1 % significance level. All variables are monthly, and the sample period is indicated for each series.

Table 3
Description of forecasting model.

Description	Model name	Equation
Benchmark model	AR	Eq. (4)
Univariate predictive model	AR-BR ⁱ	Eq. (5)
	AR-BR ^{PCA}	Eq. (6), (7)
Dimensionality reduction model	AR-BR ^{sPCA}	Eq. (8), (9), (10)
	AR-BR ^{PLS}	Eq. (11), (12), (13)

Notes: This table shows all the predictive models employed in this paper.

more stable results and offered improved predictive ability across several agricultural markets. For instance, the MEAN method generated out-of-sample R^2 values of 4.621 % ($p < 0.01$) for corn, 3.015 % ($p < 0.05$) for Soybean No. 2, and 0.617 % ($p < 0.05$) for wheat, with concomitant low MSPE. This indicates that both equal-weighted and weighted combination strategies effectively integrate multisource information and enhance forecasting performance. Further, dimension-reduction-based forecasting models, using PCA, sPCA, and PLS to

synthesize the four Biodiversity Risk series, demonstrated stable and significant predictive power across multiple contracts. In particular, the PLS-based model achieved the most comprehensive and significant forecasting results for volatility in the corn, wheat, cotton, and palm oil futures markets.

Table 5 reports the out-of-sample predictive performance of the Geopolitical Risk (GPR), Economic Policy Uncertainty (EPU), Trade Policy Uncertainty (TPU), Chinese climate policy uncertainty (CCPU), and China Economic Policy Uncertainty risk factors (CNEPU). The benchmark model used is AR-BR^{PLS}. The results show that, compared with the AR-BR^{PLS} model, the uncertainty and geopolitical risk indicators generally exhibit weaker predictive power for volatility in major agricultural futures markets. However, they perform relatively well in the case of two soybean futures contracts, consistent with our prior expectations. These findings further reinforce the predictive value of biodiversity-related risk factors in agricultural futures markets.

Overall, the constructed Biodiversity Risk indices outperformed the benchmark model in forecasting accuracy, with the PLS-based approach

Table 4Out-of-sample R^2 results of individual predictors.

Model	Corn		Soybean no.1		Soybean no.2	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-BR ^{GM}	12.058**	2.228	1.300	0.798	5.620***	3.311
AR-BR ^{ECO}	0.362	0.535	-0.634	-0.635	3.240**	1.758
AR-BR ^{PEO}	-0.258	-0.819	0.320	0.579	-1.131	-0.409
AR-BR ^{CCTV}	-1.248*	1.308	1.614*	1.372	1.520	0.969
f_MEAN	4.621***	2.592	0.951*	1.327	3.015**	2.234
f_MEDIAN	1.333**	1.679	0.854	1.226	1.976**	1.659
f_TMC	1.333**	1.679	0.854	1.226	1.976**	1.659
f_DMSPE100	4.463***	2.447	0.929*	1.302	3.044**	2.263
f_DMSPE90	4.474***	2.437	0.932*	1.306	3.062**	2.277
AR-BR ^{PCA}	4.548**	1.984	1.128	1.211	5.741***	2.336
AR-BR ^{sPCA}	4.062***	2.399	2.241*	1.286	4.705**	2.302
AR-BR ^{PLS}	4.079**	1.859	-3.222	-0.404	-3.518	-1.679
Model	Common wheat		Cotton		Palm oil	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-BR ^{GM}	-0.108	-0.531	1.561*	1.528	3.159*	1.416
AR-BR ^{ECO}	2.309***	2.500	0.623	0.976	-1.451	-0.607
AR-BR ^{PEO}	-0.078	-2.795	-1.643	-0.813	0.560	0.663
AR-BR ^{CCTV}	0.291	0.402	-0.451	-1.440	-0.154	-0.315
f_MEAN	0.617**	2.012	0.214	0.582	0.696***	2.644
f_MEDIAN	-0.096	-0.662	0.561**	2.029	0.645**	1.989
f_TMC	-0.096	-0.662	0.561**	2.029	0.645**	1.989
f_DMSPE100	0.622**	2.020	0.225	0.611	0.698***	2.543
f_DMSPE90	0.625**	2.022	0.219	0.596	0.684***	2.535
AR-BR ^{PCA}	0.065	0.223	-2.253	-1.619	-1.061	-1.715
AR-BR ^{sPCA}	1.992**	2.004	-1.825	-1.008	2.016	1.010
AR-BR ^{PLS}	1.465***	2.748	3.897**	1.873	7.686***	2.801

Notes: This table reports the results of the out-of-sample R^2 test and MSPE from AR models incorporating BR predictors derived from four different sources. Forecast combination methods (MEAN, MEDIAN, TMC, DMSPE100, DMSPE90) and dimension reduction techniques (PCA, sPCA, PLS) are also employed. A positive R^2_{Oos} indicates improved predictive performance over the historical average benchmark. The significance of the R^2_{Oos} statistic is based on [Clark and West's \(2007\)](#) MSFE-adjusted statistic, with $p < 0.1$, $p < 0.05$, and $p < 0.01$.

Table 5

Comparison with uncertainty variables.

Model	Corn		Soybean no.1		Soybean no.2	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-TPU	-4.662	-0.364	6.223	1.197	5.121**	2.143
AR-EPU	-0.928	0.559	16.145**	2.250	6.418***	2.341
AR-CNEPU	-7.648	-0.448	2.312	0.788	5.346***	2.362
AR-GPR	-2.353	0.525	9.409**	1.894	2.681	1.133
Model	Common wheat		Cotton		Palm oil	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-TPU	-1.773	-2.835	-4.264	-1.310	-8.528	-1.954
AR-EPU	-1.540	-2.743	-5.832	-1.577	-8.704	-1.763
AR-CNEPU	-1.505	-2.869	-4.114	-0.845	-9.862	-1.696
AR-GPR	-0.756	-0.409	-4.626	-1.569	-8.830	-1.649

Notes: This table reports out-of-sample R^2 and MSPE results for AR models using Trade Policy Uncertainty(TPU), Economic Policy Uncertainty(EPU), China Economic Policy Uncertainty(CNEPU), Geopolitical Risk(GPR), and Chinese climate policy uncertainty(CCPU) as single predictors. A positive value of R^2_{Oos} indicates superior forecasting performance compared to the AR-BRPLS model. The significance of the R^2_{Oos} statistic is based on [Clark and West's \(2007\)](#) MSFE-adjusted statistic, with $p < 0.1$, $p < 0.05$, and $p < 0.01$.

proving particularly well-suited for delivering stable predictions across multiple markets. This superior performance likely stems from the ability of PLS to extract latent factors that maximize covariance with the dependent variable through dimensionality reduction. By doing so, PLS effectively integrates information from multiple sources while filtering out noise, intrinsically mitigates multicollinearity, and guards against overfitting, thereby achieving the most stable and optimal predictive results in the context of a limited sample.

4.3. Forecast encompassing tests

We employ the forecast encompassing test to examine whether the AR-BR^{PLS} model, identified as having the strongest predictive performance in the previous section, provides additional forecasting information relative to other models across different futures markets. The forecast encompassing test is specified as follows:

$$RV_{t+1} = (1 - \lambda)\widehat{RV}_{q,t+1} + \lambda\widehat{RV}_{p,t+1}, \quad (16)$$

where $\widehat{RV}_{p,t+1}$ denotes the forecast from model p using the newly constructed BR, and $\widehat{RV}_{q,t+1}$ denotes the forecast from the comparison model q . If $\lambda(1 - \lambda) = 0$, an encompassing relationship holds between the two models. Specifically, when $\lambda = 0$, it implies that the predictions made by model q entirely incorporate those of model p . Conversely, when $1 - \lambda = 0$, model p contains all the information in model q . We employ the statistic proposed by [Harvey, Leybourne, and Newbold \(1998\)](#) to test the null hypothesis $H_0: \lambda(1 - \lambda) = 0$. A significant rejection of H_0 indicates that the factor-augmented model provides incremental predictive information beyond the benchmark model, demonstrating independent explanatory power and forecasting contribution.

Table 6 reports the p -values from the Forecast Encompassing Tests. The results indicate that in the wheat, cotton, and palm oil markets, the AR-BR^{PLS} model consistently provides additional predictive information beyond all competing models. In the corn market, although the AR-BR^{PLS} model is encompassed by the AR-BR^{GM} model, there is no significant evidence that the benchmark model encompasses AR-BR^{PLS}, suggesting that AR-BR^{PLS} still contributes distinct predictive content. In the soybean market, while several models are found to encompass AR-BR^{PLS}, the model nevertheless retains unique forecasting value relative to other specifications. Overall, the evidence underscores that AR-BR^{PLS} not only achieves strong standalone forecasting performance but also

Table 6

Encompassing test result.

Model	AR-BR ^{PLS}					
	Corn		Soybean no.1		Soybean no.2	
	λ	$1 - \lambda$	λ	$1 - \lambda$	λ	$1 - \lambda$
AR-BR ^{GM}	0.587	0.036	0.184	0.655	0.045	0.948
AR-BR ^{ECO}	0.044	0.518	0.089	0.839	0.001	0.995
AR-BR ^{PEO}	0.578	0.049	0.234	0.607	0.009	0.835
AR-BR ^{CCTV}	0.634	0.029	0.176	0.679	0.155	0.692
f_MEAN	0.750	0.061	0.088	0.720	0.055	0.903
f_MEDIAN	0.274	0.365	0.131	0.762	0.010	0.979
f_TMC	0.525	0.094	0.143	0.733	0.020	0.966
f_DMSPE100	0.525	0.094	0.143	0.733	0.020	0.966
f_DMSPE90	0.277	0.347	0.132	0.760	0.010	0.980
AR-BR ^{PCA}	0.276	0.347	0.132	0.760	0.010	0.980
AR-BR ^{sPCA}	0.240	0.304	0.140	0.728	0.010	0.955
AR-BR ^{PLS}	0.250	0.287	0.104	0.831	0.006	0.971

Model	AR-BR ^{PLS}					
	Common wheat		Cotton		Palm oil	
	λ	$1 - \lambda$	λ	$1 - \lambda$	λ	$1 - \lambda$
AR-BR ^{GM}	0.996	0.005	0.888	0.035	0.965	0.004
AR-BR ^{ECO}	0.991	0.008	0.816	0.058	0.986	0.000
AR-BR ^{PEO}	0.095	0.897	0.760	0.063	0.929	0.020
AR-BR ^{CCTV}	0.996	0.004	0.961	0.006	0.917	0.010
f_MEAN	0.895	0.090	0.905	0.035	0.978	0.002
f_MEDIAN	0.971	0.029	0.900	0.032	0.961	0.005
f_TMC	0.995	0.005	0.851	0.053	0.957	0.005
f_DMSPE100	0.995	0.005	0.851	0.053	0.957	0.005
f_DMSPE90	0.970	0.029	0.900	0.032	0.962	0.005
AR-BR ^{PCA}	0.970	0.029	0.900	0.032	0.962	0.005
AR-BR ^{sPCA}	0.984	0.015	0.948	0.009	0.961	0.004
AR-BR ^{PLS}	0.227	0.755	0.954	0.008	0.988	0.000

Note: The table represents p -values for [Harvey et al.'s \(1998\)](#) statistic. The p -values in the λ column examine the null hypothesis that the competing forecasts in the first column encompass the forecasts in the first row, against the alternative hypothesis that the competing forecasts in the first column do not encompass the forecasts in the first row. Similarly, the p -values in the $1 - \lambda$ column examine the null hypothesis that the forecasts in the first row encompass competing forecasts in the first column, against the alternative hypothesis that the forecasts in the first row do not encompass the competing forecasts in the first column.

delivers non-redundant information that is not fully captured by either benchmark models or alternative biodiversity risk specifications.

5. Various robustness tests

5.1. Rolling window forecasting method

To examine the robustness of our findings, we employ a rolling window forecasting approach. Specifically, we construct a fixed-size training window that moves forward over time. At each step, the model is re-estimated using the most recent observations within the window and used to forecast the target variable in the subsequent period. This method mimics real-time prediction and allows for time-varying relationships between variables.

[Table 7](#) reports the forecasting results based on a rolling time window, used to assess the temporal robustness of the main models. Overall, the R^2_{OOS} values from the rolling forecasts are consistent with the main results presented in [Table 4](#), indicating that the proposed Biodiversity Risk indices exhibit strong predictive robustness. Additionally, the PLS-based model yields a positive, though less statistically significant, forecasting coefficient in the cotton market. This outcome may be attributed to the fact that PLS relies more heavily on the full-sample structure when extracting robust factors related to price volatility, and the changing sample structure induced by the rolling training window may reduce its feature extraction efficiency.

5.2. Forecasting with a 3-year out-of-sample period

To test the robustness of our baseline results to the choice of the out-of-sample forecast horizon, we re-estimate the models using a 3-year out-of-sample period instead of the original 4-year setting. As reported in [Table 8](#), the forecasting results remain largely consistent, with the key predictors exhibiting stable R^2_{OOS} and MSPE values under the shorter horizon. This confirms that our main findings are not driven by the specific choice of the forecast window length.

5.3. Substitution of biodiversity factor

We employ an alternative biodiversity indicator constructed as follows:

$$BR2_{i,t} = 100 \times \left(\frac{\text{Negative}_{i,t} - \text{Positive}_{i,t}}{\text{Negative}_{i,t} + \text{Positive}_{i,t}} + 1 \right) \times \frac{1}{2} \times \frac{\text{BIO}_{i,t}}{\text{Length}_{i,t}}, \quad (17)$$

where $\text{BIO}_{i,t}$ represents the frequency of biodiversity-related terms in the collected articles. As shown in [Table 9](#), the forecasting results using this alternative biodiversity factor remain largely positive for corn and wheat. That said, the forecasting performance for the cotton market weakens when this alternative factor is applied. Nonetheless, the AR-BR^{PLS} model continues to outperform both the benchmark and competing models in the palm oil market.

6. Further analysis

6.1. Out-of-sample performance under high volatility regimes

In this section, we evaluate the forecasting performance of the BR index during episodes of elevated market volatility. Adopting the framework of [Liang, Tang, Li, and Wei \(2020\)](#) and [Tang, Xiao, Wahab, and Ma \(2021\)](#), we partition the out-of-sample window into high- and low-volatility regimes by comparing each month's realized volatility to its sample mean. Formally, a high-volatility state is identified when $RV_t > \overline{RV}_t$.

[Table 10](#) reports the forecasting results under a high-volatility regime. Compared to the baseline findings in [Table 4](#), the predictive performance of biodiversity risk indices improves notably during periods of heightened market uncertainty. For example, among the individual biodiversity indicators, BR^{GM} achieves an R^2_{OOS} of 21.927 % in the corn market, statistically significant at the 1 % level. However, its predictive performance appears less consistent across markets, performing poorly in the wheat market. In contrast, BR^{ECO} yields an R^2_{OOS} of 36.566 % in the wheat market, significant at the 5 % level, though its performance in other markets remains limited. Overall, models incorporating biodiversity risk measures exhibit particularly strong forecasting ability for corn and wheat futures during high-volatility periods, while their predictive power for cotton and palm oil remains broadly in line with the results reported in [Table 4](#). Notably, models based on dimensionality reduction techniques, especially PLS, continue to produce stable and superior forecasts. These findings underscore the relevance of biodiversity-related risks in predicting agricultural futures market volatility under turbulent conditions.

6.2. Multiple-steps-ahead forecasting horizon

Beyond the one-month-ahead forecasts, we also examine longer horizons, as investors typically value extended projections. Following [Zhang, Ma, and Liao \(2020\)](#), we generate three-, six-, and nine-month volatility forecasts, which are defined by the h -step-ahead aggregation: $RV_{t+h} = \frac{1}{h}(RV_{t+1} + RV_{t+2} + \dots + RV_{t+h})$.

The results in [Table 11](#) indicate that the predictive power of the biodiversity risk (BR) factor grows with longer forecast horizons in

Table 7

Out-of-sample results for rolling window.

Model	Corn		Soybean no.1		Soybean no.2	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-BR ^{GM}	11.692**	2.180	0.201	0.359	4.028***	2.628
AR-BR ^{ECO}	−1.215	−0.918	−1.843	−0.272	−0.610	0.525
AR-BR ^{PEO}	0.343	0.851	0.441	0.662	−1.209	−0.456
AR-BR ^{CCTV}	0.399*	1.536	1.934*	1.385	2.301	0.920
f MEAN	4.354***	2.429	0.797	0.968	1.726	1.203
f MEDIAN	1.109*	1.493	0.917*	1.358	0.731	0.731
f TMC	1.109*	1.493	0.917*	1.358	0.731	0.731
f DMSPE100	4.292**	2.317	0.771	0.946	1.743	1.224
f DMSPE90	4.293**	2.299	0.777	0.959	1.734	1.223
AR-BR ^{PCA}	2.107	1.179	0.378	0.458	3.419*	1.388
AR-BR ^{sPCA}	8.524**	2.171	2.280	1.096	2.424	1.273
AR-BR ^{PLS}	10.166***	2.507	−3.312	−0.170	−4.552	−2.215

Model	Common wheat		Cotton		Palm oil	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-BR ^{GM}	−0.033	−0.761	2.023*	1.455	2.319	1.101
AR-BR ^{ECO}	2.052**	2.242	1.359*	1.457	−2.895	−0.214
AR-BR ^{PEO}	0.147*	1.452	−0.946	−0.367	1.697	1.106
AR-BR ^{CCTV}	−0.145	−0.800	−0.358	−1.142	−0.053	0.015
f MEAN	0.514**	2.218	0.861	1.270	0.835	1.192
f MEDIAN	−0.004	−0.067	0.846*	1.558	1.352**	1.735
f TMC	−0.004	−0.067	0.846*	1.558	1.352**	1.735
f DMSPE100	0.518**	2.219	0.876*	1.282	0.785	1.134
f DMSPE90	0.521**	2.213	0.874	1.279	0.761	1.101
AR-BR ^{PCA}	0.001	0.018	−1.633*	−1.102	−1.286	−0.475
AR-BR ^{sPCA}	1.803**	2.032	−0.234	0.041	−1.523	−0.637
AR-BR ^{PLS}	2.201**	2.235	4.214	1.595	6.402***	3.347

Notes: This table reports the out-of-sample R^2 test results and MSPE based on a rolling window.**Table 8**

Out-of-sample results for 3-year period.

Model	Corn		Soybean no.1		Soybean no.2	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-BR ^{GM}	8.826**	1.793	0.284	0.359	4.736***	2.416
AR-BR ^{ECO}	−0.002	0.207	−0.988	−0.435	3.592**	1.698
AR-BR ^{PEO}	0.139	0.529	−1.083	−0.751	−2.328	−1.075
AR-BR ^{CCTV}	−4.549	0.799	1.929	1.027	1.664	0.909
f MEAN	3.183**	1.878	0.648	0.576	2.685**	1.725
f MEDIAN	0.439	0.611	0.383	0.440	1.740	1.274
f TMC	0.439	0.611	0.383	0.440	1.740	1.274
f DMSPE100	2.756**	1.656	0.602	0.578	2.720**	1.754
f DMSPE90	2.639*	1.622	0.594	0.573	2.738**	1.765
AR-BR ^{PCA}	1.793	0.888	−0.346	0.013	4.102*	1.551
AR-B ^{sPCA}	1.710**	1.725	0.013	0.303	4.198**	1.817
AR-BR ^{PLS}	8.210***	2.465	−0.285	0.288	−0.902	−0.692

Model	Common wheat		Cotton		Palm oil	
	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE	$R^2_{Oos}(\%)$	MSPE
AR-BR ^{GM}	−0.105	−0.524	2.105*	1.611	2.845	1.180
AR-BR ^{ECO}	2.307***	2.546	0.183	0.520	−1.446	−0.557
AR-BR ^{PEO}	−0.080	−2.938	−0.279	−0.031	0.908	0.946
AR-BR ^{CCTV}	0.288	0.394	−0.318	−0.943	0.095	0.273
f MEAN	0.616**	2.027	0.600**	1.653	0.762***	2.907
f MEDIAN	−0.098	−0.673	0.603**	1.777	0.665**	1.928
f TMC	−0.098	−0.673	0.603**	1.777	0.665**	1.928
f DMSPE100	0.619**	2.027	0.584*	1.608	0.786***	2.881
f DMSPE90	0.622**	2.033	0.578*	1.598	0.780***	2.882
AR-BR ^{PCA}	0.063	0.211	−0.954	−1.224	−0.560	−0.901
AR-BR ^{sPCA}	1.989**	2.020	−0.585	−0.521	2.352	1.081
AR-BR ^{PLS}	1.465***	2.814	4.525**	1.954	6.888**	2.139

Notes: This table reports 3-year out-of-sample R^2 and MSPE results.

Table 9

Out-of-sample results for the alternative biodiversity indicator.

Model	Corn		Soybean no.1		Soybean no.2	
	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE
AR-BR ^{GM}	8.025***	2.551	-0.056	0.211	5.980***	4.076
AR-BR ^{ECO}	-0.262	-0.706	0.009	0.142	3.626**	2.208
AR-BR ^{PEO}	2.469*	1.618	-0.810	-0.885	4.961***	2.898
AR-BR ^{CCTV}	1.412*	1.319	0.274	0.657	0.927*	1.597
f MEAN	6.423***	2.473	-0.062	-0.136	4.392***	3.627
f MEDIAN	2.395**	2.001	0.049	0.273	4.136***	3.644
f TMC	2.395**	2.001	0.049	0.273	4.136***	3.644
f DMSPE100	6.779***	2.521	-0.065	-0.148	4.408***	3.651
f DMSPE90	6.588***	2.474	-0.066	-0.149	4.423***	3.657
AR-BR ^{PCA}	4.994**	2.140	-0.275	-0.642	6.049***	3.617
AR-BR ^{sPCA}	9.492***	2.539	-0.807	-0.804	5.828***	3.433
AR-BR ^{PLS}	13.408**	2.174	-3.933	-1.033	-1.584	-0.426

Model	Common wheat		Cotton		Palm oil	
	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE
AR-BR ^{GM}	4.177***	2.430	-0.567	-0.533	-0.100	-0.078
AR-BR ^{ECO}	6.315**	1.717	0.881	0.846	-0.684	-0.320
AR-BR ^{PEO}	-0.038	-0.311	-0.778	-0.691	-0.803	-0.217
AR-BR ^{CCTV}	-0.350	-1.346	-0.338	-0.201	-1.031	-1.307
f MEAN	2.670**	2.180	-0.099	0.054	-0.543	-0.529
f MEDIAN	1.087**	2.242	-0.349	-0.279	-0.468	-0.533
f TMC	1.087**	2.242	-0.349	-0.279	-0.468	-0.533
f DMSPE100	2.667**	2.159	-0.067	0.086	-0.552	-0.541
f DMSPE90	2.704**	2.177	-0.069	0.082	-0.547	-0.535
AR-BR ^{PCA}	1.934**	2.222	-0.803	-0.275	-1.676	-1.309
AR-BR ^{sPCA}	4.031**	2.312	-0.801	-0.365	-1.734	-1.800
AR-BR ^{PLS}	2.482***	2.339	0.468	0.969	4.970***	2.489

Notes: Table 9 reports the out-of-sample R^2 and MSPE results from robustness checks using alternative biodiversity risk indicators.**Table 10**

Out-of-sample performance under the high volatility regimes.

Model	Corn		Soybean no.1		Soybean no.2	
	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE
AR-BR ^{GM}	21.927***	4.041	2.134	1.013	4.945***	2.854
AR-BR ^{ECO}	1.165	1.198	-0.866	-0.866	3.683*	1.411
AR-BR ^{PEO}	-0.264	-0.858	0.129	0.251	-0.858	-0.324
AR-BR ^{CCTV}	-1.197	1.272	1.484	1.148	1.197	0.663
f MEAN	6.858***	4.481	0.972	1.266	2.538**	1.772
f MEDIAN	2.144***	2.790	0.712	0.948	1.697*	1.353
f TMC	2.144***	2.790	0.712	0.948	1.697*	1.353
f DMSPE100	7.092***	4.649	0.950	1.242	2.559**	1.793
f DMSPE90	7.125***	4.633	0.958	1.252	2.581**	1.808
AR-BR ^{PCA}	7.170***	3.067	1.606*	1.546	5.174**	1.930
AR-BR ^{sPCA}	7.967***	3.531	3.181*	1.610	4.118**	1.890
AR-BR ^{PLS}	5.510**	2.071	-4.121	-0.634	-3.613	-1.692

Model	Common wheat		Cotton		Palm oil	
	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE	$R^2_{OOS}(\%)$	MSPE
AR-BR ^{GM}	-26.282	-0.743	1.827**	1.743	2.818	1.234
AR-BR ^{ECO}	36.566**	1.853	0.632	0.797	-1.079	-0.414
AR-BR ^{PEO}	12.993	0.977	-1.498	-0.780	0.444	0.569
AR-BR ^{CCTV}	15.758*	1.308	-0.482	-1.548	0.059	0.152
f MEAN	23.617*	1.514	0.225	0.568	0.661***	2.713
f MEDIAN	19.149	1.267	0.555**	2.033	0.786**	2.273
f TMC	19.149	1.267	0.555**	2.033	0.786**	2.273
f DMSPE100	21.906**	1.690	0.239	0.602	0.655***	2.603
f DMSPE90	22.432**	1.666	0.235	0.594	0.642***	2.593
AR-BR ^{PCA}	34.417*	1.347	-2.503	-1.901	-0.903	-1.448
AR-BR ^{sPCA}	19.117*	1.493	-1.826	-1.047	1.748	0.853
AR-BR ^{PLS}	11.315	1.192	4.896**	2.191	7.298***	2.775

Notes: This table presents the out-of-sample R^2 test under high volatility regimes. We classify volatility into high and low volatility using the mean of stock volatility in the out-of-sample interval. If $RV_t > \bar{RV}_t$, volatility is defined as high volatility; otherwise, it is defined as low volatility.

Table 11
Out-of-sample R^2 tests with different horizons.

Model	Corn	Soybean no.1	Soybean no.2	Common wheat	Cotton	Palm oil
Panel A: h = 3						
AR-BR ^{GM}	14.090***	3.417*	11.048***	−0.003	1.373	2.142*
AR-BR ^{ECO}	1.283	−0.655	6.048***	2.250***	−0.142	−0.241
AR-BR ^{PEO}	−0.141	−0.332	−0.283	0.253**	0.723	0.529
AR-BR ^{CCTV}	−0.846	4.538**	5.240*	0.261	−0.126	−0.017
f_MEAN	5.903***	2.186**	6.590***	0.697***	0.649**	0.788***
f_MEDIAN	1.805	0.504	3.619***	0.186**	0.283*	0.480**
f_TMC	1.805	0.504	3.619***	0.186**	0.283*	0.480**
f_DMSPE100	6.766***	2.198**	6.721***	0.700***	0.651**	0.799***
f_DMSPE90	6.832***	2.209**	6.778***	0.701***	0.649**	0.797***
AR-BR ^{PCA}	6.739**	2.184**	8.059***	0.237*	−0.228	−0.506
AR-BR ^{SPCA}	5.154**	4.785**	12.015***	1.767**	−0.139	1.474
AR-BR ^{PLS}	1.583	−1.969	6.513***	2.862**	5.476***	6.257***
Panel B: h = 6						
AR-BRGM	27.860***	15.205***	14.677***	−0.008	2.376	4.942**
AR-BRECO	2.616**	−0.796	4.149***	0.721*	−0.210	−0.708
AR-BRPEO	0.062	−0.108	−0.644	0.391**	1.574*	0.646
AR-BRCCTV	5.819*	4.998**	3.742*	0.371	2.211**	0.653
f_MEAN	12.418***	5.829***	6.204***	0.372**	2.200***	1.715***
f_MEDIAN	3.742**	1.424*	2.678**	0.153**	0.404	0.693**
f_TMC	3.742**	1.424*	2.678**	0.153**	0.404	0.693**
f_DMSPE100	14.299***	6.211***	6.657***	0.371**	2.113***	1.807***
f_DMSPE90	14.488***	6.362***	6.681***	0.372**	2.122***	1.816***
AR-BR ^{PCA}	12.896***	5.517***	7.007***	0.055*	0.176	−0.100
AR-BR ^{SPCA}	19.682***	12.277***	13.523***	0.426	0.109	4.022**
AR-BR ^{PLS}	12.632***	8.438**	9.324***	2.000*	8.958***	9.125***
Panel C: h = 9						
AR-BRGM	34.407***	17.557***	15.906***	−0.012	7.592***	6.630***
AR-BRECO	1.154	−0.022	1.419*	−0.011	0.748	−0.666
AR-BRPEO	−0.087	−0.278	−0.130	0.238*	0.602	0.894
AR-BRCCTV	4.215**	3.192**	2.730*	0.985	6.747**	0.268
f_MEAN	14.451***	6.205***	5.545***	0.303	5.753***	2.253***
f_MEDIAN	3.481*	1.615**	1.873**	0.048	1.727**	0.774***
f_TMC	3.481**	1.615**	1.873**	0.048	1.727**	0.774***
f_DMSPE100	17.736***	6.690***	6.095***	0.303	5.812***	2.454***
f_DMSPE90	18.218***	6.880***	6.109***	0.303	5.807***	2.489***
AR-BR ^{PCA}	13.383***	6.123***	4.579***	−0.012	3.082*	−0.152
AR-BR ^{SPCA}	23.261***	13.392***	14.903***	0.878	7.504**	4.757**
AR-BR ^{PLS}	15.733***	9.267**	14.713***	1.778	13.500***	12.332***

Notes: This table presents the out-of-sample performance of three-, six- and nine-month volatilities. A positive R^2_{OOS} value indicates that the predictive model is superior to the benchmark model.

agricultural futures volatility. In particular, BR performs significantly better at medium- (6-month) and long-term (9-month) horizons than at the short-term (1-month), suggesting that it captures persistent, forward-looking environmental risk signals suited to modeling long-term volatility. These findings both confirm the effectiveness of the BR factor and highlight its potential utility in agricultural financial markets. While single-source BR indices can deliver strong forecasts in specific markets and horizons, their performance lacks consistency. In contrast, models based on forecast combinations and dimensionality reduction methods produce more stable and superior results across all horizons. Notably, in forecasting 6-month and 9-month volatilities, the AR-BR^{PLS} model achieves the best overall performance across multiple markets, demonstrating its competitiveness and its ability to capture the long-term effects of biodiversity-related risks.

7. Conclusion

Our findings integrate biodiversity risk, providing a novel perspective for volatility forecasting in agricultural futures markets. Incorporating biodiversity risk markedly improves volatility predictions, especially for the markets of corn, wheat, cotton, and palm oil, which are the focus of this study. We constructed a biodiversity risk (BR) index utilizing four news outlets and integrated it into an autoregressive (AR)

forecasting model. Furthermore, we performed a comparison analysis of models constructed using five ensemble forecasting approaches and three dimensionality reduction techniques (PCA, SPCA, and PLS), emphasizing their out-of-sample predictive performance. Initially, out-of-sample results suggest that the developed biodiversity risk (BR) factor exhibits considerable predictive capability. The AR BR^{PLS} model demonstrates consistently robust and steady predictive accuracy when evaluating the performance of all forecasting models across various markets. Nonetheless, the predictive performance of the BR index is relatively inferior in the soybean market. To acquire a clearer comprehension of this market-specific tendency, we benchmarked AR-BR^{PLS} against the Geopolitical Risk Index (GPR) and several widely used uncertainty indices. The analysis indicates that, although uncertainty and geopolitical risk measures are more effective in predicting the soybean market, they are inferior to the AR BR^{PLS} model in all other agricultural commodity markets. The Forecast Encompassing Tests indicate that AR BR^{PLS} offers supplementary and non-nested predictive insights compared to competing models, significantly enhancing volatility forecasting in agricultural futures markets. These findings are further corroborated by a variety of robustness checks. Secondly, we assess the performance of the BR index during phases of elevated market volatility. The results demonstrate that the index's predictive ability increases during heightened market volatility. Among the evaluated models, AR

BR^{PLS} consistently provides dependable and accurate predictions across various markets in unpredictable settings. Thirdly, we broaden our study to encompass medium- and long-term horizons, concentrating on forecasts for three, six, and nine months. The BR index exhibits robust predictive performance over extended durations. Single-source BR indices sometimes exhibit inconsistent findings across markets, but models utilizing mixed sources and dimensionality reduction techniques, especially those employing Partial Least Squares (PLS), yield more stable and resilient results. AR-BR^{PLS} distinguishes itself by generating robust long-term forecasts in the corn, wheat, cotton, and palm oil markets, attributable to its proficient capacity to extract informative features from the data.

In conclusion, the biodiversity risk index significantly enhances the volatility forecasting in agricultural futures markets. This conclusion remains valid across various robustness checks, high-volatility conditions, and extended forecasting horizons. Moreover, our findings underscore the significance of forecast combining and dimensionality reduction techniques, with PLS demonstrating notably robust and consistent performance across various markets.

From an economic perspective, integrating the Biodiversity Risk Index (BR) into volatility forecasting models provides a more accurate instrument for risk evaluation among diverse market participants. Producers and hedgers can refine their hedging techniques and position sizes to mitigate risk-associated expenses, while speculators and asset managers may gain from more dependable Value-at-Risk (VaR) models and risk thresholds to improve capital allocation efficiency. Regulatory agencies may utilize BR signals as an innovative early warning indicator, promptly modifying margin requirements and liquidity restrictions during periods of increased market volatility to mitigate systemic risk. The study offers empirical evidence for the advancement of sustainable finance and ESG-related derivatives, emphasizing the significance of environmental information in agricultural credit valuation and derivative innovation. It establishes a robust basis for the synchronized progress of ecological conservation and commercial efficiency.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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